

Statistics Practice Set — S23

Probability Fundamentals

S23 starts the **Probability** section. Focus on understanding the difference between **experiment, outcome, event, and sample space** rather than memorizing definitions.

Q1 — Identify Experiment, Outcome & Event

A company randomly selects one customer from its database and checks whether the customer makes a purchase within 7 days.

Answer:

1. What is the **experiment**?
 2. Give one possible **outcome**.
 3. Define an **event** for this experiment.
 4. Is “customer makes a purchase” an outcome or an event? Explain.
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Q2 — Sample Space

A fair six-sided die is rolled once.

Answer:

1. Write the complete sample space.
 2. How many possible outcomes are there?
 3. Define the event **A = getting an even number**.
 4. Define the event **B = getting a number greater than 4**.
 5. How many outcomes are in event A?
 6. How many outcomes are in event B?
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Q3 — Event Classification

A standard deck of 52 cards is used and one card is selected randomly.

Classify each event as certain, impossible, or possible:

1. Getting a card from the deck.
 2. Getting a red card.
 3. Getting a black card.
 4. Getting a green card.
 5. Getting a card with a number greater than 20.
 6. Getting either a King or a Queen.
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Q4 — Basic Probability

A box contains:

Red

Red

Red

Blue

Blue

Green

Green

Green

Green

Yellow

One ball is selected randomly.

Answer:

1. Total number of possible outcomes.
 2. Probability of selecting Red.
 3. Probability of selecting Blue.
 4. Probability of selecting Green.
 5. Probability of selecting Yellow.
 6. Probability of **not** selecting Red.
 7. Which color has the highest probability?
 8. Which color has the lowest probability?
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Q5 — Probability from a Business Dataset

A company surveyed 100 customers about whether they purchased a premium subscription.

Customer Type	Number of Customers
Premium	35
Standard	65

One customer is randomly selected.

Answer:

1. Probability that the customer is Premium.
 2. Probability that the customer is Standard.
 3. Probability that the customer is **not Premium**.
 4. Are Premium and Standard mutually exclusive for this classification?
 5. What is the probability that the selected customer belongs to either Premium or Standard?
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Q6 — Sample Space + Event

Two coins are tossed simultaneously.

Answer:

1. Write the complete sample space.
 2. How many possible outcomes are there?
 3. Define event A = **exactly one Head**.
 4. Define event B = **at least one Head**.
 5. How many outcomes belong to event A?
 6. How many outcomes belong to event B?
 7. Calculate $P(A)$.
 8. Calculate $P(B)$.
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Q7 — Types of Events

For each pair, identify whether the events are **mutually exclusive or not mutually exclusive**.

1. **Rolling a die:**
A = getting 2
B = getting 5
2. **Rolling a die:**
A = getting an even number
B = getting an odd number
3. **Drawing one card:**
A = getting a King
B = getting a red card
4. **Drawing one card:**
A = getting a Heart
B = getting a Spade

5. **Selecting one employee:**

A = employee works in Sales

B = employee works in Finance

For each one, briefly explain **why**.

Q8 — Probability Formula Application

A website receives 500 visitors in one day.

- 125 visitors purchase a product.
- 375 visitors do not purchase.

A visitor is selected randomly.

Answer:

1. Probability of purchase.
 2. Probability of no purchase.
 3. Convert both probabilities into percentages.
 4. If 1,000 similar visitors arrive, approximately how many would you expect to purchase?
 5. Is this an experimental probability or a theoretical probability? Explain.
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Q9 — Find the Incorrect Conclusion

A student says:

“The probability of an event is 0.8, so the event will definitely happen 8 times out of every 10 trials.”

Answer:

1. Is this conclusion correct?
2. What is wrong with the statement?
3. What does probability 0.8 actually tell us?

4. If the experiment is repeated many times under the same conditions, what should happen to the observed proportion?
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Q10 — Full Data Analyst Probability Case

An e-commerce company analyzes 1,000 randomly selected website visitors.

Visitor Type	Count
Purchased	180
Added to Cart but Did Not Purchase	220
Browsed Only	600

Answer:

1. Define the **experiment**.
2. Give one possible **outcome**.
3. Define event A = visitor purchased.
4. Write the sample space for the three visitor categories.
5. Calculate $P(\text{Purchased})$.
6. Calculate $P(\text{Added to Cart but Did Not Purchase})$.
7. Calculate $P(\text{Browsed Only})$.
8. Calculate $P(\text{Purchased or Added to Cart})$.
9. Calculate $P(\text{Not Purchased})$.
10. Which outcome is most probable?
11. Are the three visitor categories mutually exclusive?
12. Give one business interpretation of the probability of **Purchased or Added to Cart**.